# Sample report: S3_GSE281305_mix3

AstroGenetics: the physics of methylation, read on one sample

Specimen constructed DNA mixture · platform EPIC

**SHOWCASE, OUTSIDE THE COMMISSIONED SCOPE: a DNA mixture of nine cell types, including two cultured cell lines, fibroblasts and endothelial cells (GSE281305). The IAM-Deconvolver is commissioned on blood-cell shares; this shows what it reports on cells it was not built for. Not a clinical result.**

**Research use only. Not a diagnostic test.** Each reading is printed with its commissioning status: *commissioned* means it passed a test sealed before its data was seen; a reading outside what an instrument was tested on is still shown, and says so. The full technical record (intake, calibration, tare, noise gate, safeguards, integrity, and how to run it yourself) is in the HTML report.

## At a glance

| Instrument | Question it answers | Reading | State | Status |
|---|---|---|---|---|
| **IAM-Deconvolver** | Which cells, in what shares | **CD4 T cells 46.3 %, B cells 20.3 %, NK cells 15.3 %** | | commissioned (neutrophils, monocytes, CD4 T, CD8 T) |
| **Met-A** | How far the neutrophil's tagging level has moved | **no reading** | withheld by the array's noise check | Met-A is not yet officially commissioned on whole blood. |
| **C-Score** | Are the departures clustered or scattered | **C 1.3313 (untared)** | | This score is preliminary until the C-Score is officially commissioned on constructed DNA mixture. |
| **IAM-A** | Is each mark still held, against the sample's own reference | | | not run (no sequencing input) |

## AstroGenetics: the physics behind each reading

One law runs under every reading here: a mark held against thermal noise costs energy, and the loss of that hold shows as entropy, in bits. Each reading is a ratio to its healthy reference, so healthy reads 1. Its gauge runs from 0, the floor (below it a cell can no longer hold its identity), to saturation, the most the scale can register (every identity site fully scrambled). Both ends are derived from physics, not fitted to data.

| Instrument | What was measured | The physics |
|---|---|---|
| **Met-A** | No reading: the noise gate withheld the state on this array (noise index 0.23749 > 0.149 and no same-run tare; A printed as a number only). | A reading is given only when enough of the array's noise sites are measured to show the array is clean enough to read. |
| **C-Score** | How the departures of Met-A's identity sites cluster along the genome: the variance of the means of blocks of 10 neighboring sites over the variance of single sites, divided by the healthy median; **C = 1.331**. | A ratio of clustering, so it has no derived floor or ceiling. Healthy reads near 1; the Normal band 0.877 to 1.152 was set on 617 healthy arrays from 19 studies and held in two laboratories that had no part in setting it. The sky below shows the same departures. |

## Cell shares: IAM-Deconvolver against IAM-Atlas v2.1

| Cell | Share | Range (information) | Status |
|---|---|---|---|
| CD4 T cells | **46.3 %** | 39.6 to 53.0 % | commissioned |
| B cells | **20.3 %** | 16.6 to 24.6 % | This share is preliminary until the IAM-Deconvolver is officially commissioned on B cells. |
| NK cells | **15.3 %** | 10.8 to 21.2 % | This share is preliminary until the IAM-Deconvolver is officially commissioned on NK cells. |
| Neutrophils | **15.1 %** | 7.1 to 20.8 % | commissioned |
| Monocytes | **3.0 %** | 0.0 to 10.3 % | commissioned |
| CD8 T cells | **0.0 %** | 0.0 to 0.1 % | commissioned |

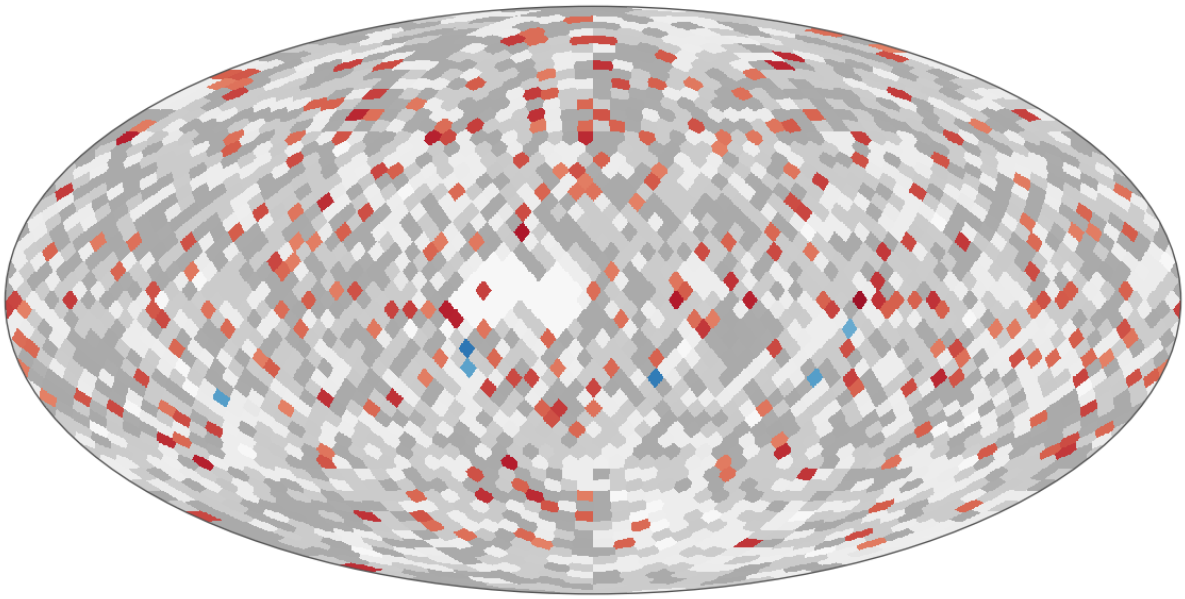
## C-Score on its bar

C-Score · constructed DNA mixture · not tared: no reading on the band



# What changed: the sample laid over its healthy reference

## What changed: this sample laid over healthy neutrophils



matches healthy (grey shade: the healthy cell's tagging)
more disorder than healthy: pattern lost (beyond +2 of its own spread)
more order than healthy (beyond −2)

C-score 1.33 · 281 of 3053 pixels changed

Grey is where this sample matches healthy neutrophils from IAM-Atlas v2.1 (the shade is the healthy cell's own tagging). Color marks only the sites that differ from healthy by more than 2 of this sample's own spread: red where its pattern is more disordered than healthy (marks lost), blue where it is more ordered. Every site keeps one place on a HEALPix sphere, set by its position in the genome, so a few scattered colored points are ordinary, and colored patches are changes that run along the genome, which is what the C-Score measures. A picture of the reading; nothing is scored from it.